



ASME V&V 40 · NASA-STD-7009B · OPEN SOURCE · APACHE-2.0

# Unit of *Assurance* (UoFA)

An open-source way to package simulation credibility evidence so a reviewer can verify it *without reading 200 pages of prose*.

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## THE PROBLEM

Reviewers don't reject your simulation. They reject your evidence package. Standards tell you *what* to assess. Your tools capture *what ran*. Neither captures *why you believe it*, and that's where submissions stall.

### C1 · PACKAGES

#### Decisions as artifacts

Your credibility evidence becomes a portable, signed object you hand a reviewer. Ed25519 + SHA-256. Tool-independent. Tamper-evident.

### C2 · DETECTS GAPS

#### A rule engine for weakeners

Forward-chaining rules catch missing UQ and unsupported claims. Surfaces compound risks no standalone SPARQL query can find.

### C3 · COMPARES

#### Two contexts of use, side by side

When the bar moves, see exactly which factor moved with it. Diff one COU against another in 30 seconds.

```
uofa rules morrison/cou2
```

```
$ uofa rules morrison/cou2.jsonld
```

#### Weakeners Detection Report

SUMMARY: 18 weakeners detected

Critical: 9  
High: 7  
Medium: 2

|               |        |        |
|---------------|--------|--------|
| ✗ W-PROV-01   | [Crit] | 7 hits |
| ✗ COMPOUND-01 | [Crit] | 2 hits |
| △ W-EP-04     | [High] | 6 hits |
| △ W-ON-02     | [High] | 1 hit  |
| △ W-AL-02     | [Med]  | 1 hit  |
| △ W-CON-04    | [Med]  | 1 hit  |

✗ 2 compound inferences — chained,  
not detectable by standalone SPARQL.

## WHY IT MIGHT FIT YOU

Works in **medical device** (ASME V&V 40) and **aerospace** (NASA-STD-7009B) today. Runs locally, so evidence never leaves your environment. Sits alongside your PLM and SPDM — it doesn't replace them.

### ZERO INSTALL

codespaces.new/  
cloudronin/uofa

### ON YOUR MACHINE

pip install uofa  
uofa setup

### WORKED EXAMPLE

uofa.net/demo



SCAN FOR DEMO

**The ask:** Run it on one of your own models — even a small one. Email me what breaks or what's missing.

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uofa.net · [github.com/cloudronin/uofa](https://github.com/cloudronin/uofa)